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Paleomagnetic and rock magnetic data from the Silurian succession of the Dniester basin, Ukraine

Maria Jeleńska[a,*], Vladimir Bakhmutov[b], Leonid Konstantinienko[c]

[a] *Institute of Geophysics, Polish Academy of Science, Ks. Janusza 64, 01-452 Warsaw, Poland*
[b] *Institute of Geophysics, Ukraine Academy of Science, Palladin Avenue 32, 252680 Kiev-142, Ukraine*
[c] *Institute of Geology, Ukraine Academy of Science, Gonchara Street 55b, Kiev 01054, Ukraine*

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Abstract

Paleomagnetic and rock magnetic data from Silurian sedimentary rocks from west Ukraine, comprised of limestones and dolomites of the Yaruga, Malinovtsy and Rukshin series of Wenlock to Downton age, have been obtained. The rock magnetic data revealed magnetite as the main magnetic carrier. The rocks are characterized by pervasive Permian and Carboniferous remagnetization, demonstrated by the Permian "P" direction, which yields a pole position of 52°S, 356°E, identified in all the rocks, and the Late Carboniferous "C" direction, only found in Rukshin series with a pole position of 29°S, 356°E. The third magnetization, "S", is considered primary or early diagenetic origin and yields a pole position of 15°S, 350°E which corresponds to the Silurian part of the APWP for Baltica.
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Keywords: Silurian; Ukraine; Paleomagnetic poles; Magnetic properties

1. Introduction

The paleocontinent of Baltica consisted of the Scandinavia and Russian Platform in the early Palaeozoic (Torsvik, 1998). The Russian Platform lies between the Urals to the east and the East European Platform to the west. The apparent polar wander path (APWP) for Baltica constructed by Torsvik et al. (1992, 1996) was widely accepted as a reference curve. The part of the path before the Carboniferous was based mainly on Scandinavian and Russian data. Most of paleomagnetic poles of Silurian and Devonian age published for the Russian Platform are coincident with the Late Palaeozoic part of the APWP. So, there is a suspicion that these poles are not well dated or are based on secondary magnetization. Consequently, the Silurian-Devonian part of the curve is based mainly on extrapolation between best quality poles (Torsvik et al., 1996). Although apparently well defined, the Silurian part lying close to the Carboniferous segment of the APWP needs verification

* Corresponding author. Tel.: +48 22 6915762; fax: +48 22 6915915.
E-mail address: bogna@igf.edu.pl (M. Jeleńska).

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by new reliable data. Smethurst and Khramov (1992) studied the Upper Silurian limestones and Devonian sandstones from the western Ukraine. They found a Permian secondary magnetization and what is interpreted as a direction of Devonian magnetization but they were not successful in recovering the Silurian magnetization. This did not discourage us from resampling the Silurian sediments in the Podolia region of western Ukraine.

2. Geological setting

Silurian strata compose the main part of the Lower Paleozoic succession of the Dniester basin, south-western Ukraine. They rest distinctly, transgressively on an erosion surface of Upper Proterozoic, Lower Cambrian, Middle and Upper Ordovician sediments and dip gently several degrees westwards and south-westwards (Tsegelnyuk et al., 1983).

Geologically, the Middle Dniester area corresponds to the Podolian edge of the crystalline basement of the south-western part of the Russian Platform. The Silurian rocks are exposed along the Dniester river that cut the sediments across the strike as a band limited by the village of Naddnestrianka in the west, where the uppermost Silurian underlies Lower Devonian strata. The width of this band is about 80 km (Fig. 1).

The attention of geologists and paleontologists has been attracted for a long time to the Silurian sedimentary rocks because of their quality exposure on the high and steep slopes of rivers and valleys, an almost horizontal position, weak metamorphism and the abundance and breadth of fossils. Three series have been established in the Dniester sequence. In stratigraphic succession they are: Yaruga, Malinovtsy and Rukshin, according to the stages of the Silurian—early Devonian sedimentation in the territory of the south-western margin of the Russian Platform (Tsegelnyuk et al., 1983). They correspond to the English Wenlockian, Ludlovian and Downtonian (Fig. 2). The time scale of Harland et al. (1990), modified using the early Palaeozoic chronology of Tucker and McKerrow (1995), is shown in the right-hand part of Fig. 2.

The series in Podolia are separated by discontinuities of sedimentation. Despite this, they form one of the most complete sections in the world (Tsegelnyuk et al., 1983). The rocks are represented by neritic (coastal)

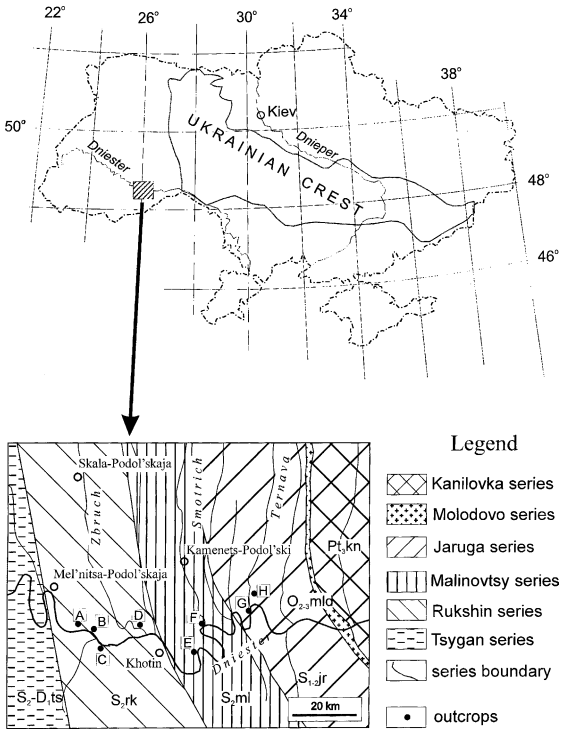


Fig. 1. Geological sketch map of the sampling area and its location in Ukraine. The letters indicate the stratigraphic sections sampled.

and pelagic (more deep-water) facies. The coastal facies are separated by lagoon (dolomites and marls), coastal (detrital and organic—detrital limestones and organic marls) and shelf (clay limestone with marl layers) facies (Malinovtsy and Rukshin series). The pelagic facies consists predominantly of clay rich limestones, marls, and argillite (Yaruga).

The base of the sequence is the Yaruga series, with total thickness of 94–135 m. The lower part of the Yaruga series is predominantly composed of marls, the middle, thickest part features limestones. Dolomites are specific for the upper part. The Malinovtsy series, with a total thickness of 90–141 m, consists predominantly of limestones. The Rukshin series is represented by limestones, dolomites and marls, its thickness varies between 170 and 200 m. Detailed sedimentary logs and more geological information are reported by Tsegelnyuk et al. (1983).

Eight main outcrops which cover all the sections except the Bagovitsa suite were sampled for paleomagnetic purposes (Fig. 1). In Fig. 2 (left-handed part),

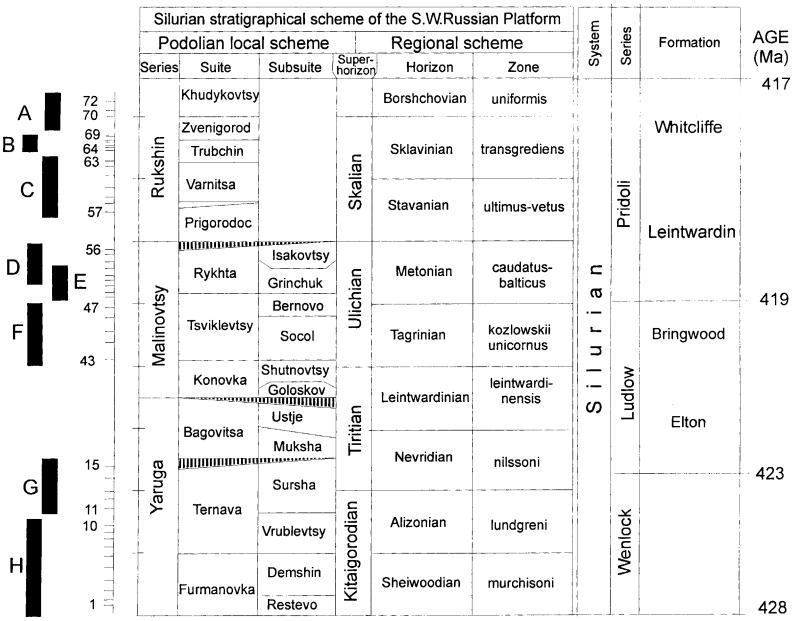


Fig. 2. Stratigraphic scheme of Silurian rocks in Podolia on the left hand, compared with the local scheme of Silurian system in Great Britain (on the right). On the right time scale of Harland et al. (1990) modified according to Tucker and McKerrow (1995). Black bars with letters on the left show stratigraphic sections with sampled levels denoted by numbers.

the sections sampled are given with the local stratigraphy scheme and the number of the level. One or two hand samples were taken from each of the 44 stratigraphic levels. Levels 1–15 come from the Yaruga series and consist of limestones. Levels 43–53 represent limestones of the Malinovtsy series. Levels 54–56 consist of dolomites. Levels 57–72 belong to limestones of the Rukshin series, except levels 62, 64, 66 and 67 which consist of dolomites. Marls were not sampled.

The preliminary results of the study were reported by Bakhmutov et al. (2001).

3. Experimental procedure

A total of 241 cubic specimens, 2 cm × 2 cm in size, were cut from hand samples (3–6 specimens from one sample). Remanent magnetization was measured with a 2G SQUID cryogenic magnetometer accompanied by AF demagnetiser. Thermal demagnetisation was performed in screened furnace made by Magnetic Measurements, UK. Paleomagnetic measurements were carried out in the magnetically shielded room. Demagnetization results were analysed using the principal component analysis (Kirschvink, 1980) by means of a PDA program package (Lewandowski et al., 1997). The KLY 2 Kappabridge was used for measurements of mean susceptibility and anisotropy of magnetic susceptibility. Anisotropy parameters were calculated using the program Aniso of Jelinek (1977).

3.1. Magnetic mineralogy

Magnetic mineralogy was determined by examination of thin and polished sections with an ore microscopy and by response to thermal treatment, including continuous thermal demagnetization of saturation remanence SIRM, thermal demagnetization of three-component IRM (Lowrie, 1990) and observation of changes of magnetic susceptibility during heating. SIRM thermal demagnetization was carried out with the use of a home-made device. A specimen was magnetized in a field of 9 T and then heated in a screen. During heating magnetization was measured in a screen by means of a modified rotation magnetometer made in the USSR. After cooling to room temperature, a specimen

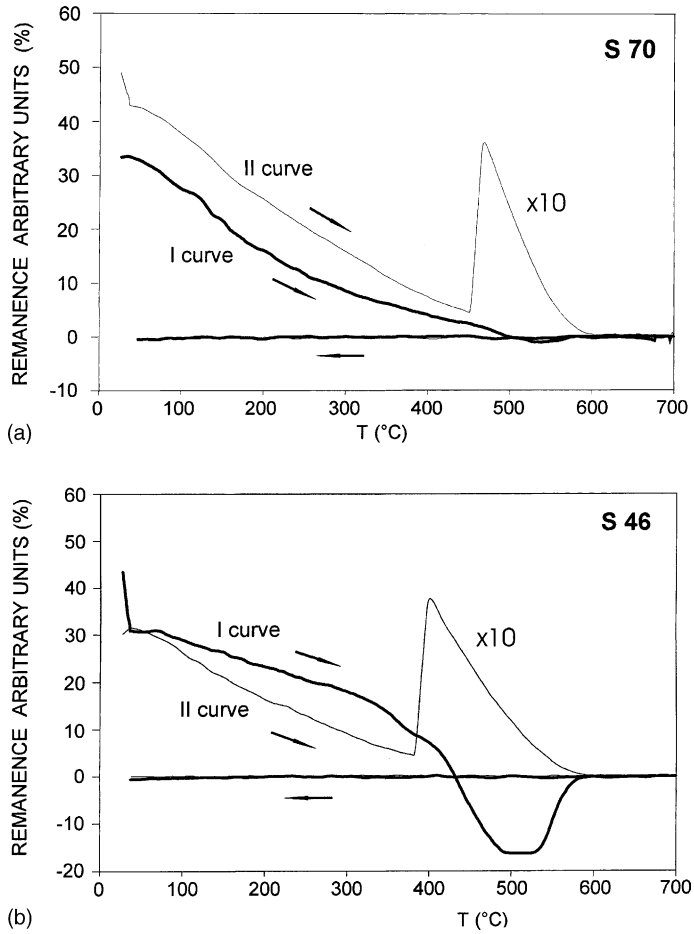


Fig. 3. The changes in saturation remanence during continuous heating: (a) sample from Rukshin series; (b) sample from Malinovtsy series. I curve: heating of fresh sample; II curve: second run for heated sample after cooling and once more magnetizing. At the end of second run the range of sensitivity was increased by 10 times.

was again magnetized and second run of measurement was performed. The components of IRM were acquired in a field of 0.11, 0.4 and 3 T in three perpendicular directions. Hysteresis parameters were measured for few specimens with a Molyneaux vibrating magnetometer VSM.

Microscopic examination of thin and polished sections was not informative. There is abundant framboidal and automorphic pyrites partly altered to iron hydro-oxides. In the Rukshin series, pyrite is preserved in small amounts mainly in calcite veins. However, it was not possible to recognize either pyrrhotite or goethite as well as magnetite and hematite. The thermomagnetic experiments were more informative. Examples of decay curves of SIRM are shown in Fig. 3. For most samples from the Rukshin series (Fig. 3a), SIRM unblocked at temperature 570 °C indicating the presence of magnetite. For most samples from the Malinovtsy and Yaruga series (Fig. 3b) SIRM increased, sometimes with a change of polarity, at temperature above 400 °C, reached the maximum at 500 °C and unblocked at 570 °C. During the second run, the increase was not observed and only magnetite was seen, which unblocked at 570 °C. Susceptibility measuremeant during continuous heating (Fig. 4a and b) also showed an increase of susceptibility after heating to 400° C. The increase is much higher and starts at a lower temperature for samples from the Malinovtsy and Yaruga

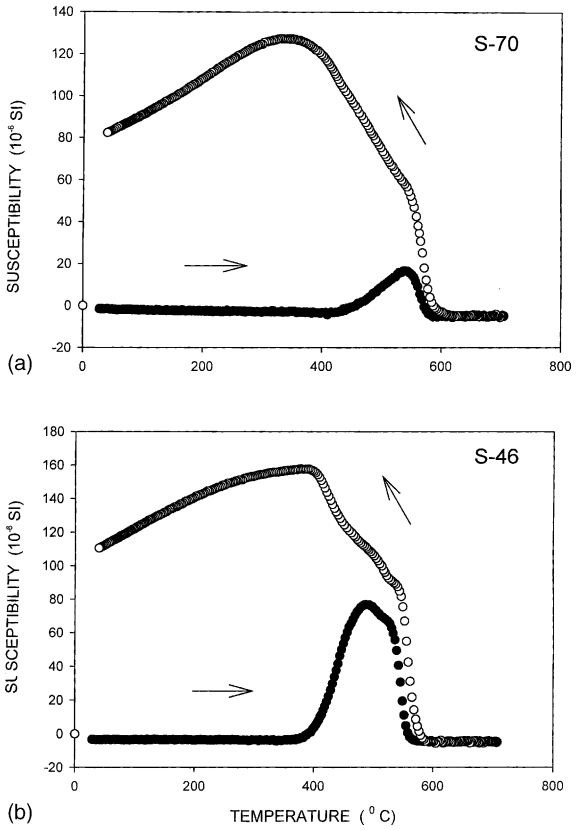


Fig. 4. Examples of changes of bulk susceptibility with temperature, measured during continuous heating: (a) sample from Rukshin series; (b) sample from Malinovtsy series.

(Fig. 4b) than for the Rukshin series (Fig. 4a). Thermal demagnetization of three-component IRM (Fig. 5a and b) showed that, in the case of the Rukshin series, the hard component demagnetized at 550 °C, the medium component at 400 °C, and the soft component between 400 and 550 °C. Although the thermomagnetic curves did not show the presence of pyrrhotite, for samples from the Malinovtsy and the Yaruga series (Fig. 5b) the soft and medium components of IRM decreased at temperature 300–350° C characteristic for phyrrotite, indicating that some amount of pyrrhotite existed (Lowrie, 1990). For the Malinovtsy and Yaruga series small amounts of hematite are visible on the curve of the hard component (Fig. 5b). IRM acquisition curves (Fig. 6a) demonstrate a steep increase of IRM related to magnetite for rocks from the Rukshin series whereas the IRM-curves for samples from the Malinovtsy and Yaruga series (Fig. 6b) exhibited two magnetic components with different acquisition behaviour. The steep increase of IRM at very low dc field strength is related to magnetite, and following a slow increase without reaching saturation, can be related to hematite which was previously observed in thermal demagnetization of the three-component IRM. This is in agreement with the hysteresis parameters. Although hysteresis parameters were measured for a few samples only, H_c and H_{cr} values showed harder magnetic material for the Malinovtsy series than for the Rukshin series (see Table 1).

Such increase of remanence as we observe during thermal demagnetisation of SIRM for samples from the Malinovtsy and Yaruga series (Fig. 3b) has been reported for greigite, pyrite and pyrrhotite (Roberts and Turner, 1993; Roberts, 1995; Ward, 1970). The

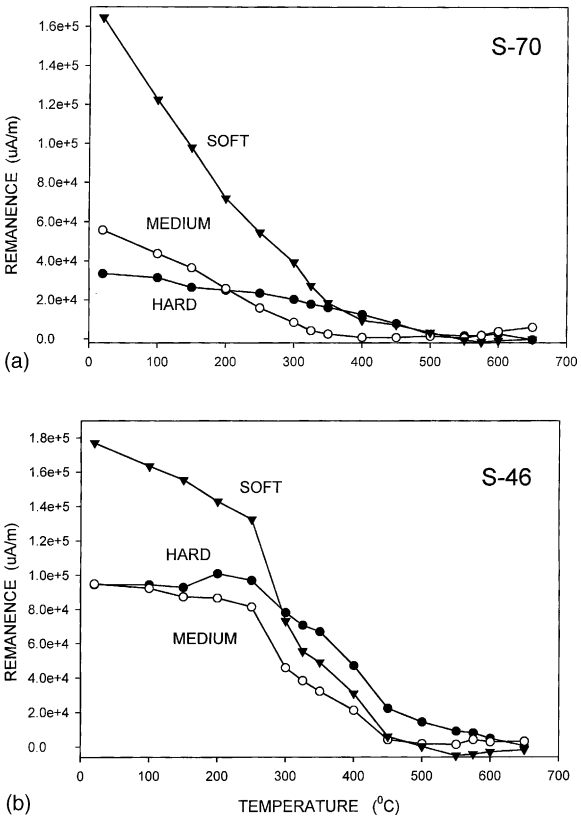


Fig. 5. Thermal demagnetization of the three-component IRM: (a) sample from Rukshin series; (b) sample from Malinovtsy series. The three perpendicular components of IRM were acquired in a field of 0.11 T (soft), 0.4 T (medium) and 3 T (hard).

Table 1
Hysteresis parameters

Series	H_c (mT)	H_{cr} (mT)	M_s (μA m^{-2})	M_r (μA m^{-2})	H_{cr}/H_c	M_r/M_s
Rukshin	6.5	55	0.82	0.12	8.46	0.15
Malinovtsy	38	80	0.77	0.11	2.1	0.14
Yaruga	Too weak for measurement					

reverse of remanence during heating of pyrrhotite was observed by Bina and Daly (1994) and Zapletal (1993). Sharp decrease of soft and medium components at temperature 300–350° C during heating of three-components IRM can indicate the presence of phyrrotite. However, we did not observe pyrrhotite on our thermomagnetic curves. This inconsistency in thermomagnetic behaviour is difficult to explain. Possibly the reason for different magnetic behaviour could be related to different time in which both experiments were performed. We can only state that the increase of remanence can be related to oxidation of sulphides to magnetite. As sulphides were formed in reducing environments (Allen and Hajek, 1989), their persistence excludes oxidation conditions. The presence of sulphides in the rocks from the Malinovtsy and Yaruga series suggests that some amount of magnetite in these rocks may be of detrital origin. As in the rocks from the Rukshin series sulphides are observed only in calcite veins, they could be at least partly oxidized to magnetite in nature.

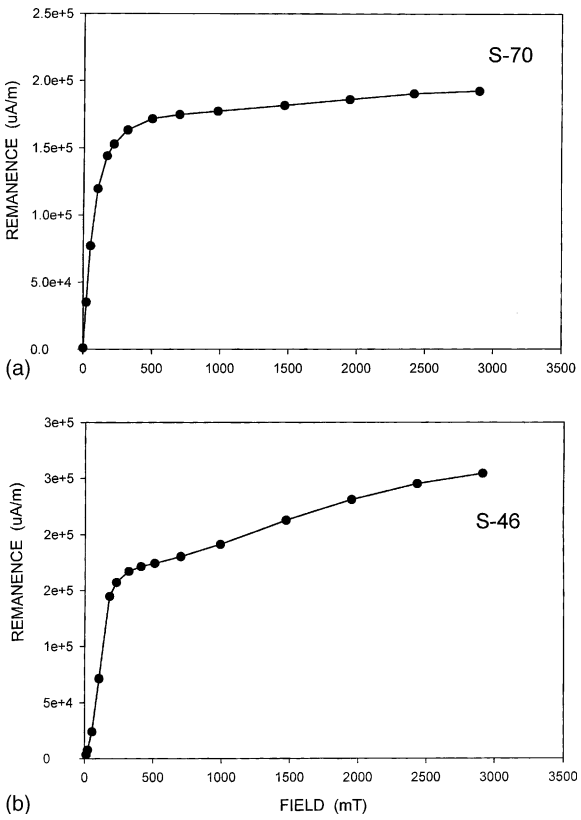


Fig. 6. IRM acquisition curves for: (a) sample from Rukshin series; (b) sample from Malinovtsy series.

3.2. Paleomagnetism

The natural remanent magnetization (NRM) ranged from 0.1 to 2.5 mA m^{-1} (Fig. 7). Mean susceptibility ranged from 2 to 168 × 10^{-6} SI (Fig. 7). The highest and the most dispersed values of NRM are associated with the youngest Rukshin series.

Thermal treatment are useful for rocks from the Rukshin series (Fig. 8a). The other rocks experience strong mineralogical alteration when heated to temperaturs above 400 °C (Fig. 8c). When thermal demagnetization was not affected by mineralogical changes, NRM was completely removed at a temperature of 470 °C. Temperature of 50° C erased the viscous component. Two components of magnetization with distinct unblocking temperatures were identified: one normal polarity component of steep northerly direction, named LB, was unblocked at temperature of 200–250 °C. The second, reverse polarity component of shallow southern direction, named HB, was unblocked between 450 and 470 °C. AF treatment up to an alternating field of 150 mT sometimes did not completely remove NRM (Fig. 8b). The LB and HB components at the beginning (below 20 mT) and above 30 mT are isolated. In six specimens, components of southerly direction with positive inclination of intermediate stability (IB) are isolated. LB and HB components have

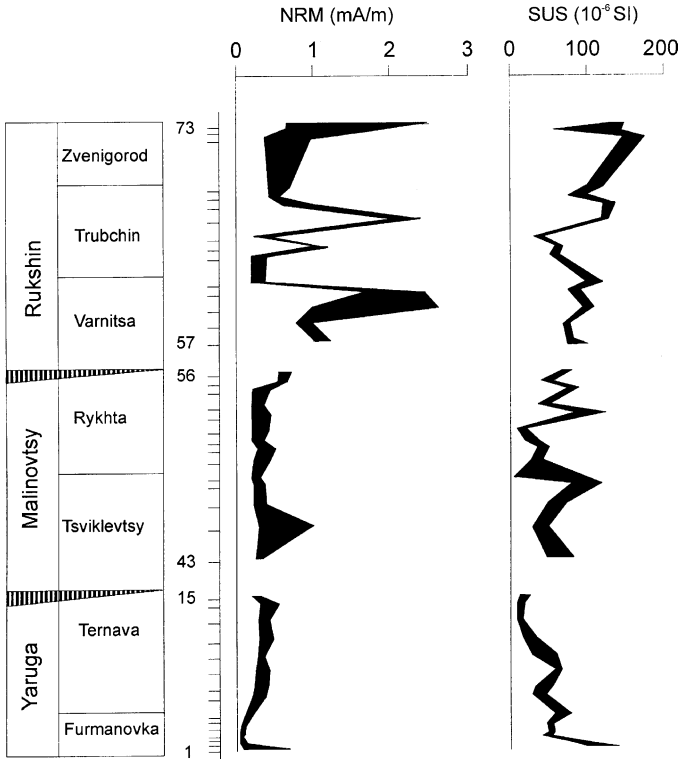


Fig. 7. Distribution of NRM intensity and bulk susceptibility along the studied sections. The band represents lowest and highest values measured within a level.

often an overlapping coercivity spectra and the demagnetization trajectory defines great circle. We have tried to apply great circles analysis (Halls, 1976; Hoffman and Day, 1978) to such data to check the possibility of isolation IB component. The intersection of planes found for the specimens from one level gave common direction, not precisely defined, close to the HB direction (Fig. 9) but with a higher inclination. It is probably that the intersection of great circles gives the direction composed of the LB and HB components. We are not successful in recovery of IB component. It seems probable that LB is not preserved in most samples from the Rukshin series.

Most samples from the Malinovtsy and Yaruga series were demagnetized by AF treatment, because the rocks experience strong mineralogical alteration when heated above 350 °C (Fig. 8c). Thermal demagnetization of pilot samples up to 350 °C revealed only recent or viscous remagnetization. During AF demagnetisation, two or three components of magnetization were identified (Fig. 8d–f). The first component, removed below 20 mT, is similar to the LB magnetization found in the Rukshin series. The HB component, removed between 30 and 160 mT, was identified in many but not all samples. In these rocks, a magnetization similar in direction to the IB component found in the Rukshin series, of intermediate or high coercivity, was identified. This direction was identifies in more than half of the samples. Occasionally, it is a second component removed after erasing the LB component to the end of the demagnetization treatment (Fig. 8d and e), sometimes it is a component of intermediate stability removed between removal of the LB and HB components (Fig. 8f). The high coercivity of some rocks from the Malinovtsy and Yaruga series did not allow for complete demagnetization of NRM even up to 160 mT. The component of very high stability, similar to the LB component (named VHB), was identified in 20 specimens from the Malinovtsy series and in five specimens from the Yaruga series.

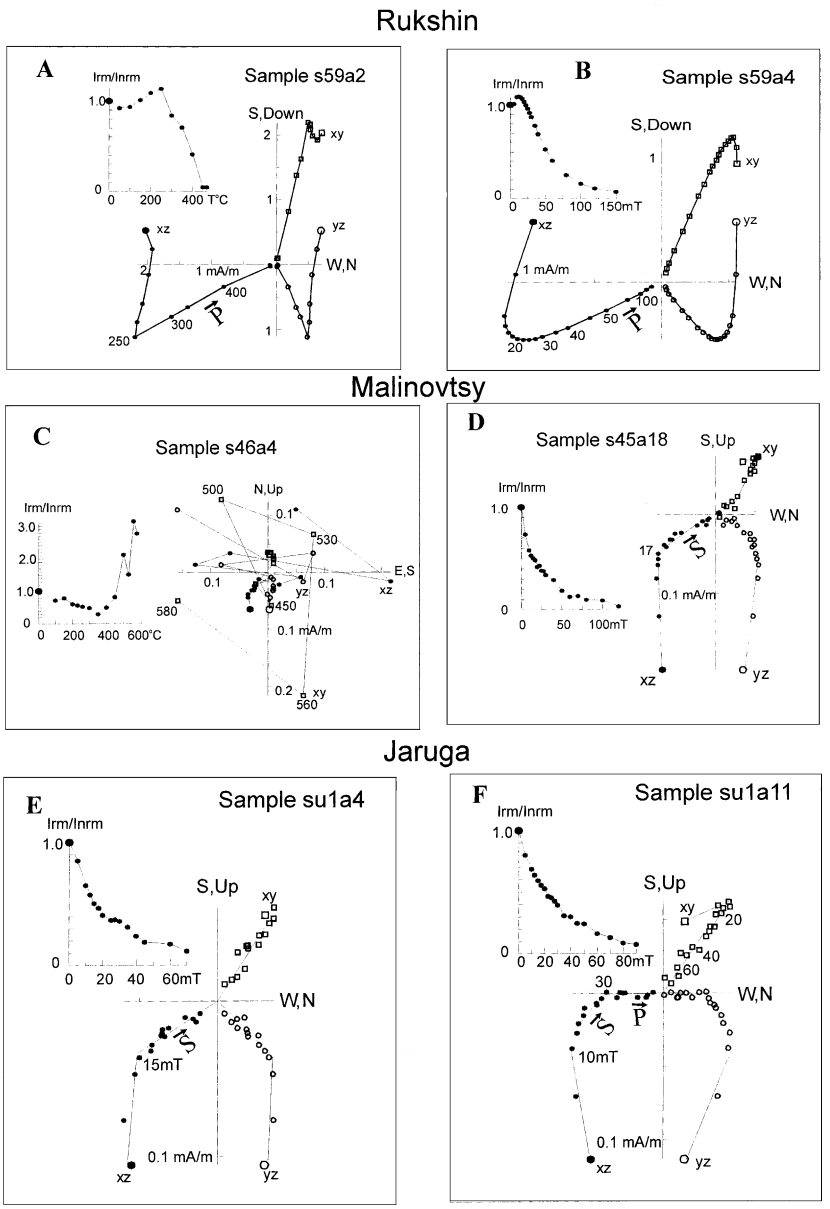


Fig. 8. Examples of thermal and AF demagnetization behaviour of samples from Rukshin (A and B), Malinovtsy (C and D) and Yaruga (E and F) series. Orthogonal diagrams are shown in horizontal plane *xy* represented by squares, and in two vertical planes: *xz* represented by solid circles and *yz* represented by open circles. Arrows with letters indicate the components isolated. The plots in upper left corners show decay of NRM intensity during demagnetisation.

4. Interpretation

The characteristic magnetizations identified in the Rukshin, Malinovtsy and Yaruga series are displayed in Fig. 10 and listed in Table 2. The mean directions calculated for the series are shown in Fig. 11. The low stability (LB) and very high stability (VHB) directions are regarded as young secondary magnetization and they were not considered further.

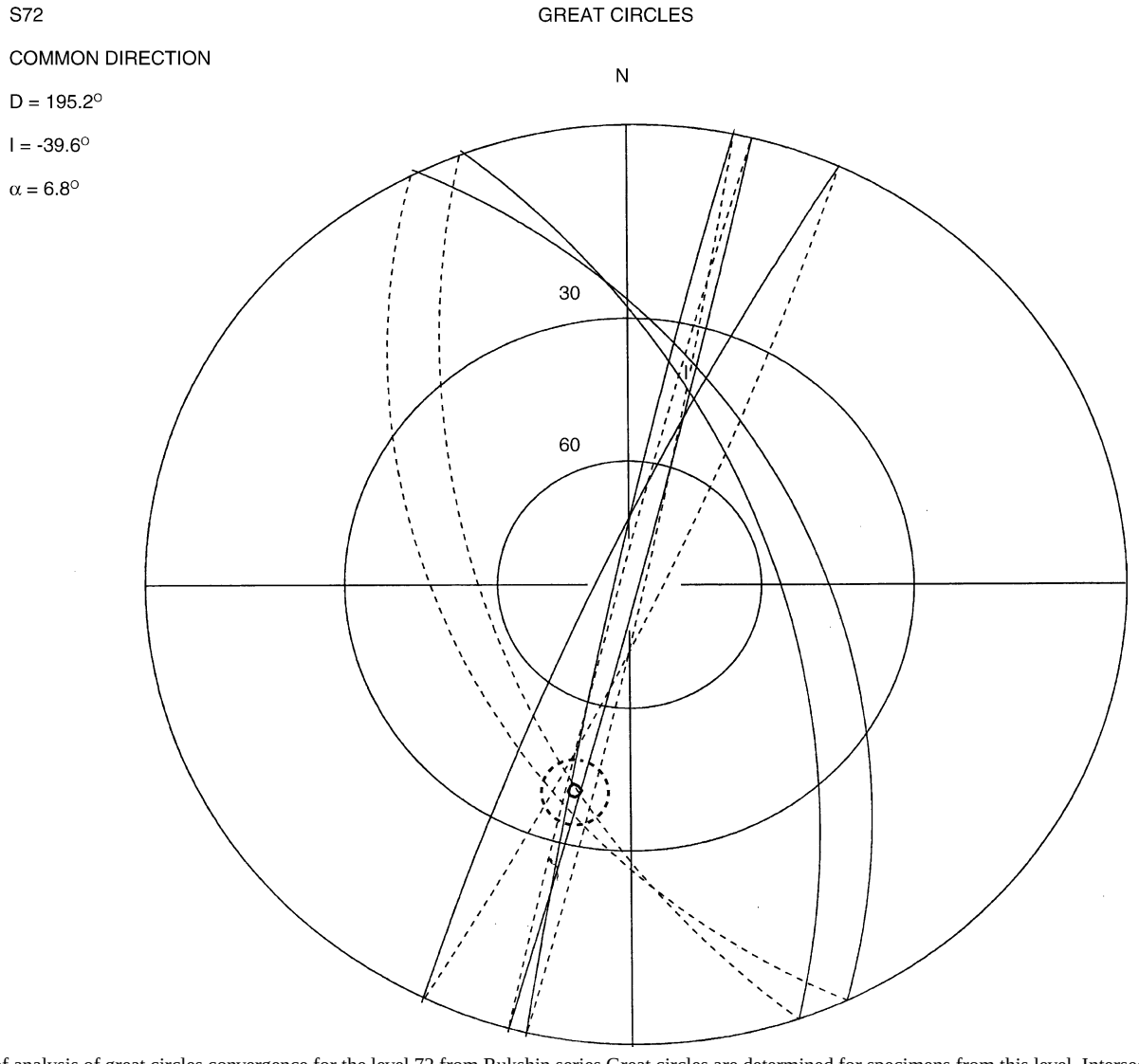


Fig. 9. Examples of analysis of great circles convergence for the level 72 from Rukshin series Great circles are determined for specimens from this level. Intersection of the circles gives the direction $D = 15.2^{\circ}$; $I = 39.6^{\circ}$; $\alpha = 6.8^{\circ}$.

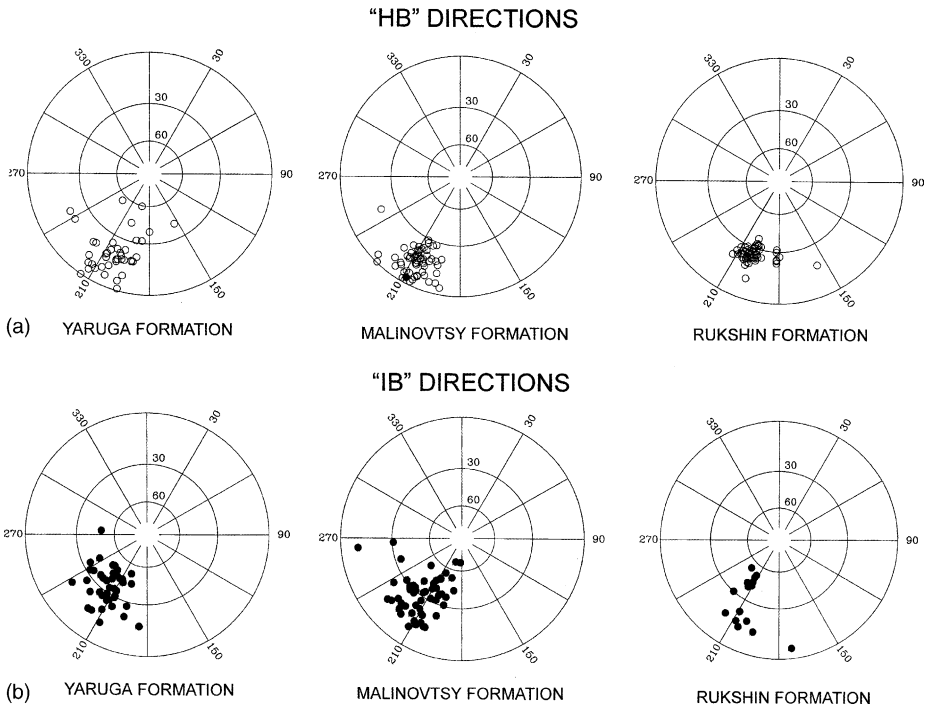


Fig. 10. Equal area projection of sample-mean directions for the formations: (a) HB directions; (b) IB directions. Solid (open) circles indicate lower (upper) hemisphere projection.

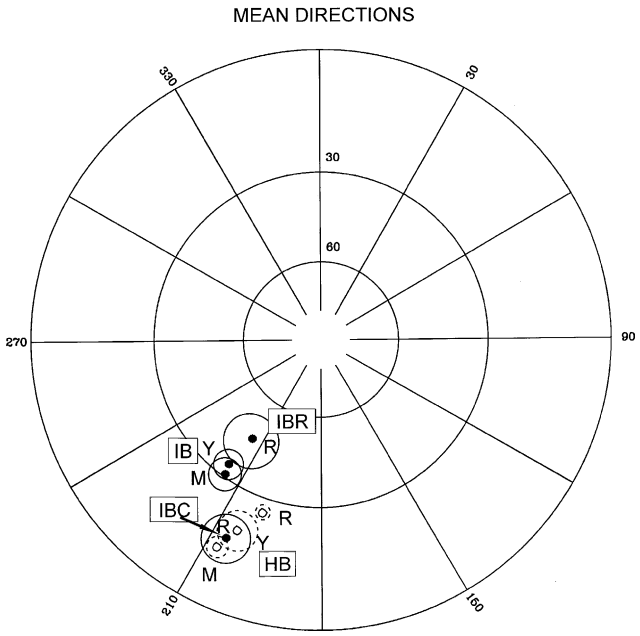


Fig. 11. Equal area projection of mean HB and IB directions for the formations shown with 95% confidence circles. Solid (open) circles indicate lower (upper) hemisphere. Y: data for Yaruga series; M: data for Malinovtsy series; R: data for Ruksin series.

Table 2
Mean characteristic directions for the Podolia area

Direction	Series	N/n	D	I	k	α_{95}	Lat	Long
Recent LB, VHB	Rukshin	61/55	345	82	25.8	3.8	63	18
	Malinovtsy	97/76	335	80	23.6	3.4	65	6
	Yaruga	83/65	334	84	13.6	5	59	16
	Malinovtsy	97/20	333	72	35	5.6	71	335
	Yaruga	83/5	360	73	131.5	6.7	80	25
Permian HB	Rukshin level	61/61	199	−26	72.6	2.1	−52	356
		15/15	198	−26	90.9	4	−52	356
	Malinovtsy level	97/60	207	−13	51.4	2.6	−42	349
		14/13	208	−16	98.8	4.2	−43	347
	Yaruga level	83/41	204	−19	18.6	5.3	−47	351
		15/11	206	−19	26.6	9	−45	348
Carboniferous IBC	Rukshin level	61/6	206	16	118.8	6.2	−29	356
		15/4	207	21	210.5	6.3	−26	356
Silurian IBR	Rukshin level	61/9	215	45	106.5	5	−9	354
		15/2	219	42			−9	350
Silurian IB	Malinovtsy level	97/52	216	31	16.9	4.9	−17	350
		14/11	215	30	32.5	8.1	−18	350
	Yaruga level	83/42	217	34	23.6	4.6	−15	350
		15/8	219	32	37.1	9.2	−16	348

N is the number of specimens measured; *n* is the number of specimens used to compute the mean value. In the case of level statistics (the lower data for Permian, Carboniferous and Silurian directions) *N* is the number of level sampled; *n* is the number of level for which mean direction was calculated. *D* (°), *I* (°), *k* and α_{95} (°) are respectively the values of declination, inclination and statistical parameters (Fisher, 1953) associated with the mean value. Lat (°), Long (°) are latitude and longitude of the pole in the southern hemisphere. The upper values for Permian, Carboniferous and Silurian directions represent specimen means; the lower values represent level means.

The HB component observed in all the sites is carried by magnetite of high coercivity, as it was removed at a temperature of about 470 °C or AF range between 30 and 160 mT. The HB component is extremely well defined in the Rukshin series and relatively poorly defined in the Yaruga series (Fig. 10). The mean directions of this component for the Rukshin, Malinovtsy and Yaruga series are: $D = 199°$, $I = -26°$; $D = 207°$, $I = -13°$, $D = 204°$, $I = -19°$, respectively (Table 2). They lie close to the secondary Permian direction ($D = 206$, $I = -20$) identifies in Devonian and Silurian sedimentary rocks from Podolia by Smethurst and Khramov (1992).

The IB component is found mainly in the Malinovtsy and Yaruga series ($D = 216$, $I = 31$; $D = 217$, $I = 34$, respectively). In the Rukshin series the group of reverse polarity magnetizations with positive inclination can be divided into the subgroup of directions with inclination higher than 30° (named IBR) and the subgroup of directions with inclination lower than 30° (named IBC). The significance test (McFadden and Lowes, 1981) determined for the two subgroups showed that they are different at the 99.9% level of significance. The IBR is isolated in nine specimens from two levels ($D = 215$, $I = 45$). The IBC was found in six specimens from four levels. We regarded the IBC as magnetic overprint. When compared with the APWP for Baltica (Torsvik et al., 1992), a Late Carboniferous age is indicated ($D = 206$, $I = 16$). In the Malinovtsy series the IB is found in 52 specimens from 11 levels, and in the Yaruga series—in 42 specimens from eight levels (Table 2). The maximum density for the IB in these series was found to be slightly above the inclination of 30°. The stability of this component is intermediate. The IB component is usually removed by a field of 50–60 mT and is isolated by AF treatment only. As magnetite is the main magnetic carrier in our rocks, the IB component must be carried by magnetite of intermediate coercivity. As thermomagnetic experiments described previously revealed the presence of sulphides,

suggesting that reducing conditions have lasted at least in part of the rock history, the preservation of primary magnetite is possible. As magnetite produced by oxidation is usually fine-grained and magnetically hard (Zegers et al., 2003), it is more likely that the primary magnetite is of intermediate coercivity.

Measurements of the anisotropy of magnetic susceptibility show that the sedimentary fabric was preserved in rocks from the Malinovtsy and Yaruga series because the foliation plane lies almost flat.

On this basis, the IB direction is interpreted as being primary, of Silurian age. There is some controversy about the polarity of the IB direction. Our data are of exclusively reverse polarity. The Silurian data set compilation given by Trench et al. (1993) demonstrated normal polarity of the paleomagnetic field during Wenlock and mixed polarity during Ludlow and Pridoli time. The rocks studied are confined to the Pridoli, Ludlow and the uppermost Wenlock strata. However, we did not sample the continuous stratigraphic sequence as it was not available in the Dniester basin, and the Silurian magnetostratigraphy is not well defined.

The IB direction is close to the Silurian direction expected for the Podolia region, based on the APWP of Baltica (Torsvik et al., 1992). The IB poles calculated for the Malinovtsy and Yaruga series (17S, 350E; 15S, 350E) fall within the Silurian poles of the age between 428 and 420 Ma of the APWP for Baltica (Fig. 12). This supports a Silurian age of the IB directions. The IB pole from the Rukshin series lies north of the other poles. It is possible that the Silurian component in this series is not well isolated and contaminated by younger

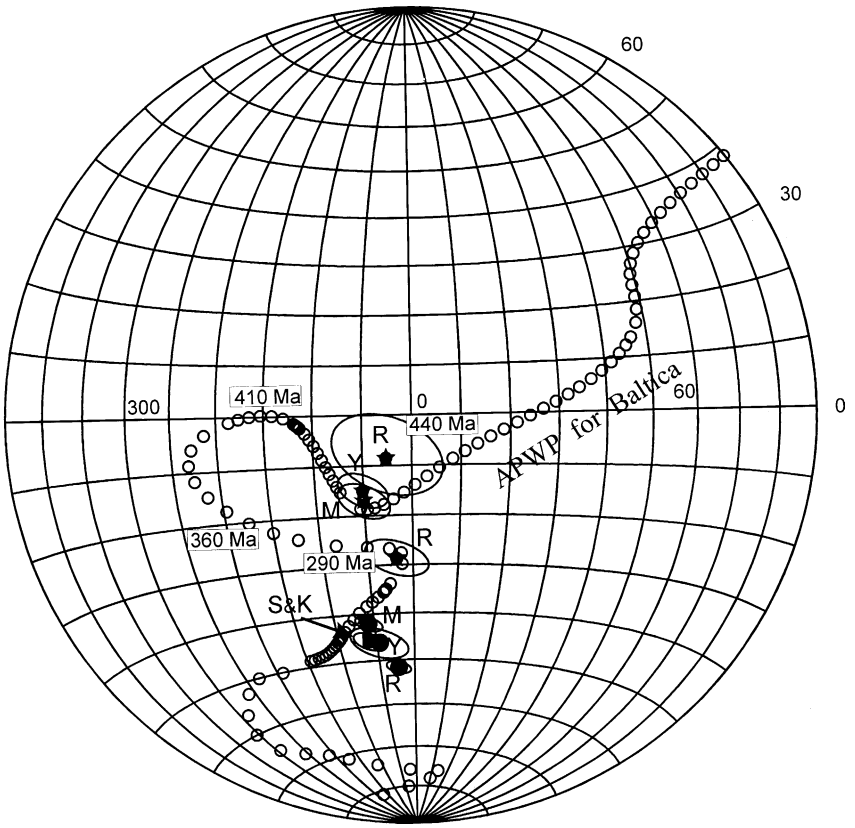


Fig. 12. Reference APWP for Baltica on equal area projection (after Torsvik et al., 1992) along with Permian (HB), Carboniferous (IBC) and Silurian (IB, IBR) poles for Podolia sediments shown with error ovals. Stars: Silurian poles; circles: Permian poles; square (S&K): pole calculated from secondary magnetization component obtained by Smethurst and Khramov (1992) for Silurian and Devonian sedimentary rocks from Podolia. The other notation as in Fig. 11.

magnetization. The rocks from the Rukshin series are the most altered ones and the isolation of the primary direction is difficult.

5. Conclusions

Silurian sediments from Podolia revealed four components of magnetization: a recent one of low and very high stability, a Carboniferous to Permian overprint of high stability, and a Silurian primary magnetization of intermediate stability. The Permian, Carboniferous and Silurian directions of magnetization were carried by magnetite of different coercivity and probably of different origin. The Carboniferous and Permian directions are carried by the finest grains of high coercivity and of secondary origin. The Silurian directions are carried by magnetite of moderate coercivity and are considered detrital in origin.

The Carboniferous direction yields a pole position 29S, 356E, in agreement with the Baltica pole position for 300 Ma. The Permian poles from Podolia sediments lie slightly apart from the APWP for Baltica. Such a deviation was observed by Jeleńska et al. (2003) for the Permian poles from the Polish Sudetes.

The Silurian poles for West Ukraine are consistent with the Silurian data for Baltica. The results from Ukraine are important, as the Silurian segment of the APWP for Baltica is poorly defined because of the lack of reliable data of this age for tectonically stable parts of Europe.

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